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St. Joseph's College of Arts & Science (Autonomous)

St. Joseph's College Road, Cuddalore – 607001

B.Com (BM) DEGREE EXAMINATION APRIL 2021

ASCM301T - BUSINESS STATISTICS

Time : 3 hrs **Max Marks :75**

SECTION - A (10x2=20)

Answer ALL the questions

1. Explain the term tabulation.
2. Define classification.
3. Find the range of weight of 10 students from the following.

   41 11 14 15 73 64 54 45 36 20
4. Explain Karl Person's co-efficient of skewness.
5. Define regression lines.
6. Write the various types of correlation.
7. What is an index number?
8. What do you meant by seasonal variation?
9. What is meant by statistical quality control?
10. What are the various types of control charts known to you?

SECTION - A (5 x 5 =25)

Answer **ALL** the questions

11. (a). Calculate Harmonic mean for the following data

Size of Items	6	7	8	9	10	11
frequency	4	6	9	5	2	8

(Or)

(b) Explain the various methods of collecting primary data?

12. (a). Calculate Quartile deviation and its Co-efficient.

Age in years	20	30	40	50	60	70	80
No. of members	3	61	132	153	140	51	3

(Or)

(b) Calculate mean deviation from and its co-efficient from the following data

X:	2	4	6	8	10
f:	1	4	6	4	1

13. (a) From the following data calculate the co-efficient of rank correlation between x and y.

X	36	56	20	65	42	33	44	50	15	60
Y	50	35	70	25	58	75	60	45	80	38

(Or)

(b) Given the bivariate data

X	1	5	3	2	1	1	7	3
Y	6	1	0	0	1	2	1	5

Fit a regression line of y on x and hence predict y if x = 5

14. (a) The following are the group index numbers and the group weights of an average working class family's budget , construct the cost of living index number.

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Group	Food	Fuel and lighting	Clothing	Rent	Miscellaneous
Index number	352	220	230	160	190
Weight	48	10	8	12	15

(Or)

(b) Calculate trend values by the method of least squares from the data given below and estimate the sales for 2003.

Year	1996	1997	1998	1999	2000
Sales('000 000)	70	74	80	86	90

15. (a) Explain chance causes and assignable causes? Give example.

(Or)

(b) Explain briefly the construction and uses of mean chart and range chart.

SECTION - C (3 X 10 = 30)

Answer **any THREE** questions.

16. Calculate mean, median and mode for the following data.

Profits per shop(in lakhs) :	0 – 10	10 – 20	20 – 30	30 – 40	40 – 50	50 – 60
No. of Shops :	12	18	27	20	17	6

17. Calculate Standard deviation and its relative measures from the following data.

C.I	:	5-10	10-15	15-20	20-25	25-30	30-35	35-40	40-45
f	:	6	5	15	10	5	4	3	2

18. Calculate correlation co-efficient for the following data.

Advertisement (cost Rs'000)	39	65	62	90	82	75	25	98	36	78
Sales (in lakhs)	47	53	58	86	62	68	60	91	51	84

19. Calculate price index from the following by (a) simple aggregative method and (b) Average of price relative method by using both Arithmetic mean and Geometric mean.

Commodity	A	B	C	D	E	F
Price in 2012(Rs)	200	300	100	250	400	500
Price in 2013(Rs)	250	300	150	350	450	550

20. Draw the mean and range charts from the following data and state your conclusion.

Sales	1	2	3	4	5	6	7	8	9	10	Total
Sample mean	12.8	13.1	13.5	12.9	13.2	14.1	12.1	15.5	13.9	14.2	135.3
Sample Range	2.1	3.1	3.9	2.1	1.9	3.0	2.5	2.8	2.5	2.0	25.9

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